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LOS Assignment Theory-3

1

Device / Partition	Filesystem Type	Size	Mount Point
/dev/nvme0n1	NVMe Disk	476.9 GB	—
/dev/nvme0n1p1	vfat (FAT32)	260 MB	/boot/efi
/dev/nvme0n1p2	No filesystem	16 MB	—
/dev/nvme0n1p3	NTFS	164.5 GB	—
/dev/nvme0n1p4	NTFS	1000 MB	—
/dev/nvme0n1p5	Btrfs	311.2 GB	/
/dev/zram0	Swap	7.6 GB	[SWAP]

1. My system has one main storage device, /dev/nvme0n1, with a size of **476.9 GB**.
2. It has five partitions: p1, p2, p3, p4, and p5.
3. The filesystem types used are **FAT32, NTFS, and Btrfs**.
4. The root / directory is located on **/dev/nvme0n1p5**.
5. The filesystem used for / is **Btrfs**.
6. The root filesystem has **312 GB total space, 135 GB used space, and 173 GB available space**.

My Review

From the above investigation, I found that my Linux system uses the **Btrfs filesystem** for the root partition, while the system also contains FAT32 and NTFS partitions.

2

./dev/nvme0n1 (476.9 GB NVMe Disk)

|

```
|— nvme0n1p1 (260 MB) → vfat (FAT32) → /boot/efi
|— nvme0n1p2 (16 MB) → no filesystem
|— nvme0n1p3 (164.5 GB) → NTFS → not mounted
|— nvme0n1p4 (1000 MB) → NTFS → not mounted
└— nvme0n1p5 (311.2 GB) → Btrfs → /
    |— /home
    |— /root
    |— /srv
    |— /var/log
    |— /var/cache
    └— /var/tmp
```

/dev/zram0 (7.6 GB) → swap → [SWAP]

Explanation:

The disk (nvme0n1) is the physical SSD, split into partitions to organize storage. Each partition has a filesystem (Btrfs, NTFS, FAT32) that manages the files on it, and the mount point is the folder where Linux makes that filesystem accessible, like / for Btrfs and /boot/efi for FAT32.

3. Exercise 3 – Explore the Root Filesystem

Start from your current location.

1. Determine your current working directory.
2. Navigate to the root directory.
3. Display the contents of the root directory.

4. Identify all directories directly available under `/`.

5. Determine whether hidden entries are present.

```
anonymous@anonymous-supreme in ~ via v3.14.7
# ) pwd
/home/anonymous

anonymous@anonymous-supreme in ~ via v3.14.7
# ) cd /

anonymous@anonymous-supreme in /
# ) pwd
/

anonymous@anonymous-supreme in /
# ) ls
drwxr-xr-x - root 31 Aug 19:03 .snapshots
lrwxrwxrwx - root 12 Oct 2025 bin → usr/bin
drwxr-xr-x - root 31 Aug 18:17 boot
drwxr-xr-x - root 7 Sep 00:13 dev
drwxr-xr-x - root 7 Sep 00:13 etc
drwxr-xr-x - root 4 Sep 16:41 home
lrwxrwxrwx - root 12 Oct 2025 lib → usr/lib
lrwxrwxrwx - root 12 Oct 2025 lib64 → usr/lib
drwxr-xr-x - root 21 Nov 2024 mnt
drwxr-xr-x - root 19 Aug 16:02 opt
dr-xr-xr-x - root 7 Sep 00:13 proc
drwxr-xr-x - root 22 Apr 2025 root
drwxr-xr-x - root 7 Sep 00:16 run
lrwxrwxrwx - root 12 Oct 2025/sbin → usr/bin
drwxr-xr-x - root 27 Mar 2025 srv
dr-xr-xr-x - root 7 Sep 00:48 sys
drwxrwxrwt - root 7 Sep 00:34 tmp
drwxr-xr-x - root 31 Aug 18:00 usr
drwxr-xr-x - root 31 Aug 18:52 var
.rw-r--r-- 7 root 8 Mar 2025 .garuda-tools

anonymous@anonymous-supreme in /
# ) ls -la /
dr-xr-xr-x - root 23 Oct 2025 .
dr-xr-xr-x - root 23 Oct 2025 ..
drwxr-xr-x - root 31 Aug 19:03 .snapshots
lrwxrwxrwx - root 12 Oct 2025 bin → usr/bin
drwxr-xr-x - root 31 Aug 18:17 boot
drwxr-xr-x - root 7 Sep 00:13 dev
drwxr-xr-x - root 7 Sep 00:13 etc
drwxr-xr-x - root 4 Sep 16:41 home
lrwxrwxrwx - root 12 Oct 2025 lib → usr/lib
lrwxrwxrwx - root 12 Oct 2025 lib64 → usr/lib
drwxr-xr-x - root 21 Nov 2024 mnt
drwxr-xr-x - root 19 Aug 16:02 opt
dr-xr-xr-x - root 7 Sep 00:13 proc
drwxr-xr-x - root 22 Apr 2025 root
```

```
anonymous@anonymous-supreme in /
# ) find / -maxdepth 1 -type d -printf '%p\n' | sort
/
/boot
/dev
/etc
/home
/mnt
/opt
/proc
/root
/run
/.snapshots
/srv
/sys
/tmp
/usr
/var

anonymous@anonymous-supreme in /
# ) ls -la /
dr-xr-xr-x - root 23 Oct 2025 .
dr-xr-xr-x - root 23 Oct 2025 ..
drwxr-xr-x - root 31 Aug 19:03 .snapshots
lrwxrwxrwx - root 12 Oct 2025 bin → usr/bin
drwxr-xr-x - root 31 Aug 18:17 boot
drwxr-xr-x - root 7 Sep 00:13 dev
drwxr-xr-x - root 7 Sep 00:13 etc
drwxr-xr-x - root 4 Sep 16:41 home
lrwxrwxrwx - root 12 Oct 2025 lib → usr/lib
lrwxrwxrwx - root 12 Oct 2025 lib64 → usr/lib
drwxr-xr-x - root 21 Nov 2024 mnt
```

Exercise 4 – Linux Filesystem Hierarchy Investigation

Investigate the following locations:

/home

/root

/etc

/var

/var/log

/tmp

/usr

/boot

/dev

/proc

For each location:

- 1. Examine its contents.
- 2. Identify at least three entries, where available.
- 3. Determine the likely purpose of the directory.

Complete:

Directory	Example Contents	Purpose
-----	-----	-----
`/home`		
`/root`		
`/etc`		
`/var`		
`/var/log`		
`/tmp`		

```
| `usr` | | |
| `boot` | | |
| `dev` | | |
| `proc` | | |
```

Directory	Example Contents	Purpose
/home	anonymous	Contains the personal files and directories of normal users.
/root	Not accessible with normal user permissions	Home directory of the root (administrator) user.
/etc	a2ps, alsa, arp-scan, audit	Contains system-wide configuration files and settings.
/var	cache, darkstat, db, games	Stores variable data that changes during system operation.
/var/log	audit, cups, garuda, journal	Stores system, security and application log files.
/tmp	.font-unix, .ICE-unix, .X11-unix, plasmashell-RiNwSy	Stores temporary files and data created by applications and the system.
/usr	bin, include, lib, lib32	Contains user-space programs, libraries and other software resources.
/boot	efi, grub, memtest86+, initramfs-linux-zen.img	Contains files required for booting the Linux system.
/dev	block, bus, char, disk	Contains special files used to access devices and hardware.
/proc	1, 2, 3, 4	Virtual filesystem containing information about running processes and the Linux kernel.

Exercise 5 – Find Your Home Directory

Without assuming its location:

1. Determine your username.
2. Determine your home directory.
3. Navi

Exercise 6 – Hidden Files Investigation

Examine your home directory.

Compare:

ls

ls -a

Identify at least 2 hidden files/directories.

Answer:

1. How does Linux identify a hidden file?
2. Why does a normal `ls` not display these files?
3. What do `.` and `..` mean?
4. Can a hidden file still be accessed if its name is known?
5. From a cybersecurity perspective, why should hidden files not automatically be considered safe or malicious?o your home directory.
4. Display its absolute path.
5. List its normal contents.
6. List all contents, including hidden files.

```

anonymous@anonymous-supreme in /
/ ) whoami
anonymous

anonymous@anonymous-supreme in /
/ ) echo "$HOME"
/home/anonymous

anonymous@anonymous-supreme in /
/ ) getent passwd "$(whoami)" | cut -d: -f6
/home/anonymous

anonymous@anonymous-supreme in /
/ ) cd "$HOME"

anonymous@anonymous-supreme in ~ via v3.14.7
/ ) pwd
/home/anonymous

anonymous@anonymous-supreme in ~ via v3.14.7
/ ) ls
drwxr-xr-x - anonymous 30 Jul 10:14 .anaconda
drwxr-xr-x - anonymous 7 Sep 00:16 .cache
drwxr-xr-x - anonymous 31 Aug 18:17 .cargo
drwxr-xr-x - anonymous 8 Sep 2025 .conda
drwxr-xr-x - anonymous 7 Sep 00:56 .config
drwx----- - anonymous 2 Aug 11:42 .copilot
drwxr-xr-x - anonymous 2 Sep 2025 .dotnet
drwx----- - anonymous 5 Sep 2025 .firedragon
drwx----- - anonymous 18 Aug 09:40 .gnupg
drwxr-xr-x - anonymous 8 Mar 2025 .icons
drwxr-xr-x - anonymous 28 Sep 2025 .ipynb_checkpoints
drwxr-xr-x - anonymous 5 Sep 2025 .ipython
drwx----- - anonymous 7 Sep 2025 .john
drwxr-xr-x - anonymous 11 Sep 2025 .jupyter
drwxr-xr-x - anonymous 11 Sep 2025 .local
drwxr-xr-x - anonymous 8 Sep 2025 .npm
drwx----- - anonymous 5 Sep 2025 .nv
drwxr-xr-x - anonymous 8 Sep 2025 .nvm
drwx----- - anonymous 27 Mar 2025 .pki
drwxr-xr-x - anonymous 27 Mar 2025 .var
drwxr-xr-x - anonymous 6 Aug 2025 .vscode

```

```

anonymous@anonymous-supreme in ~ via v3.14.7
/ ) ls -la
drwxr-xr-x - anonymous 7 Sep 00:16 .
drwxr-xr-x - root 4 Sep 16:41 ..
drwxr-xr-x - anonymous 30 Jul 10:14 .anaconda
drwxr-xr-x - anonymous 7 Sep 00:16 .cache
drwxr-xr-x - anonymous 31 Aug 18:17 .cargo
drwxr-xr-x - anonymous 8 Sep 2025 .conda
drwxr-xr-x - anonymous 7 Sep 00:56 .config
drwx----- - anonymous 2 Aug 11:42 .copilot

```

Exercise 6 – Hidden Files Investigation

Examine your home directory.

Compare:

ls

ls -la

Identify at least 2 hidden files/directories.

Answer:

1. How does Linux identify a hidden file?
2. Why does a normal `ls` not display these files?
3. What do `.` and `..` mean?
4. Can a hidden file still be accessed if its name is known?

5. From a cybersecurity perspective, why should hidden files not automatically be considered safe or malicious?

```
anonymous@anonymous-supreme in ~ via v3.14.7
) ls -ld ~/.config ~/.cache
drwxr-xr-x - anonymous 7 Sep 00:16 /home/anonymous/.cache
drwxr-xr-x - anonymous 7 Sep 01:10 /home/anonymous/.config

anonymous@anonymous-supreme in ~ via v3.14.7
```

I used `ls` and `ls -a` in my home directory to check for hidden files and directories. I found hidden directories such as `.config` and `.cache`.

1. **How does Linux identify a hidden file?**

Linux considers a file or directory hidden if its name starts with a dot (`.`).

2. **Why does a normal `ls` not display these files?**

The normal `ls` command does not show hidden files by default. We can use `ls -a` to see them.

3. **What do `.` and `..` mean?**

`.` means the current directory, and `..` means the parent directory.

4. **Can a hidden file still be accessed if its name is known?**

Yes, a hidden file can still be accessed if we know its name and location. For example, `.config` can be accessed using `~/.config`.

5. **From a cybersecurity perspective, why should hidden files not automatically be considered safe or malicious?**

Some hidden files are normal configuration files used by applications. However, malware can also use hidden files to hide suspicious activities. So, we should check the file's contents, permissions, ownership, and purpose before deciding whether it is safe or malicious.

Exercise 7 – Build a Small Filesystem Hierarchy

Explore: `mkdir`

Hint: you can create this hierarchy in a single command

`cyber_lab/`

```
+-- evidence/
|   +-- network/
|   +-- system/
|
+-- logs/
+-- scripts/
|
+-- reports/
```

Create the following files:

cyber_lab/evidence/network/traffic.txt

cyber_lab/evidence/system/processes.txt

cyber_lab/logs/security.log

cyber_lab/scripts/check.sh

cyber_lab/reports/investigation.txt

Use appropriate Linux commands to create the structure.

Verify that the required hierarchy has been created successfully.

Hint: tree

Explore options

```
anonymous@anonymous-supreme in ~ via v3.14.7 took 2s7ms
[Q] / find ~/cyber_lab -print
/home/anonymous/cyber_lab
/home/anonymous/cyber_lab/evidence
/home/anonymous/cyber_lab/evidence/network
/home/anonymous/cyber_lab/evidence/network/traffic.txt
/home/anonymous/cyber_lab/evidence/system
/home/anonymous/cyber_lab/evidence/system/processes.txt
/home/anonymous/cyber_lab/logs
/home/anonymous/cyber_lab/logs/security.log
/home/anonymous/cyber_lab/scripts
/home/anonymous/cyber_lab/scripts/check.sh
/home/anonymous/cyber_lab/reports
/home/anonymous/cyber_lab/reports/investigation.txt

anonymous@anonymous-supreme in ~ via v3.14.7 as took 14s263ms
/ ) tree ~/cyber_lab
/home/anonymous/cyber_lab
├── evidence
│   ├── network
│   │   └── traffic.txt
│   └── system
│       └── processes.txt
├── logs
│   └── security.log
├── reports
│   └── investigation.txt
└── scripts
    └── check.sh

7 directories, 5 files

anonymous@anonymous-supreme in ~ via v3.14.7 as
```

Exercise 8 – Absolute and Relative Paths

Assume your current directory is:

~/cyber_lab

Perform the following:

1. Navigate to `evidence/network` using a relative path.
2. Navigate to `/etc` using an absolute path.
3. Return to your home directory using the shortest possible command.
4. Navigate to `cyber\_lab/reports`.
5. Move one directory upward.
6. Display the current directory after every operation.

* What is an absolute path?

* What is a relative path?

* What is the difference between `/' and `~`?

```

anonymous@anonymous-supreme in ~ via v3.14.7 took 1m34s992ms
[ ] / cd ~/cyber_lab
    pwd
/home/anonymous/cyber_lab

anonymous@anonymous-supreme in ~/cyber_lab
/ } cd evidence/network
    pwd
/home/anonymous/cyber_lab/evidence/network

anonymous@anonymous-supreme in ~/cyber_lab/evidence/network
/ } cd /etc

anonymous@anonymous-supreme in /etc
/ } pwd
/etc

anonymous@anonymous-supreme in /etc
/ } cd ~

anonymous@anonymous-supreme in ~ via v3.14.7
/ } pwd
/home/anonymous

anonymous@anonymous-supreme in ~ via v3.14.7
/ } cd cyber_lab/reports

anonymous@anonymous-supreme in ~/cyber_lab/reports
/ } cd ..

anonymous@anonymous-supreme in ~/cyber_lab
/ } pwd
/home/anonymous/cyber_lab

anonymous@anonymous-supreme in ~/cyber_lab
/ } _

```

1. What is an absolute path?

An absolute path gives the complete location of a file or directory starting from the root /. For example, /home/anonymous/cyber_lab/reports.

2. What is a relative path?

A relative path gives the location based on the current directory. For example, cyber_lab/reports is a relative path when starting from /home/anonymous.

3. What is the difference between / and ~?

/ is the **root directory** of the entire Linux filesystem. ~ represents the **home directory of the current user**; in my case, it represents /home/anonymous.

Exercise 9 – Identify File Types

Investigate the following:

/etc/passwd

/etc

/bin/bash

/dev/null

/proc

Also investigate the files created in Exercise 7.

file <filename>

ls -l <filename>

Classify each object.

Object	Type
--------	------

-----	----
-------	------

`/etc/passwd`	
---------------	--

`/etc`	
--------	--

`/bin/bash`	
-------------	--

`/dev/null`	
-------------	--

`/proc`	
---------	--

`security.log`	
----------------	--

`check.sh`	
------------	--

```
(Konsole) File Edit View Bookmarks Plugins Settings Help Search (Tue)
anonymous@anonymous-supreme in ~/cyber_lab
# ) ls -l /etc/passwd
-rw-r--r-- 2.9k root  4 Sep 16:41 /etc/passwd

anonymous@anonymous-supreme in ~/cyber_lab
# ) ls -ld /etc
drwxr-xr-x - root  8 Sep 09:24 /etc

anonymous@anonymous-supreme in ~/cyber_lab
# ) ls -l /bin/bash
-rwxr-xr-x 1.2M root 10 Jun 10:02 /bin/bash

anonymous@anonymous-supreme in ~/cyber_lab
# ) ls -l /dev/null
crw-rw-rw- 1,3 root  8 Sep 09:08 /dev/null

anonymous@anonymous-supreme in ~/cyber_lab
# ) ls -ld /proc
dr-xr-xr-x - root  8 Sep 09:08 /proc

anonymous@anonymous-supreme in ~/cyber_lab
# ) file ~/cyber_lab/logs/security.log
/home/anonymous/cyber_lab/logs/security.log: empty

anonymous@anonymous-supreme in ~/cyber_lab
# ) file ~/cyber_lab/scripts/check.sh
/home/anonymous/cyber_lab/scripts/check.sh: empty

anonymous@anonymous-supreme in ~/cyber_lab
# ) ls -l ~/cyber_lab/scripts/check.sh
-rw-r--r-- 0 anonymous  7 Sep 01:18 /home/anonymous/cyber_lab/scripts/check.sh

anonymous@anonymous-supreme in ~/cyber_lab
# ) ls -l ~/cyber_lab/scripts/security.sh
"/home/anonymous/cyber_lab/scripts/security.sh": No such file or directory (os error 2)

anonymous@anonymous-supreme in ~/cyber_lab
[*] #
```

Object	Type
/etc/passwd	Regular file (text file)
/etc	Directory
/bin/bash	Regular executable file
/dev/null	Character device file
/proc	Directory (virtual filesystem)
security.log	Empty regular file
check.sh	Empty regular file

Exercise 10 – Examine File Metadata

File Metadata

Item	Details
Filename	investigation.txt
File size	46 bytes
File type	Regular file
Owner	anonymous
Group	anonymous
Inode number	5500882
Access time	2026-09-07 01:18:44 +0530
Modification time	2026-09-08 11:23:31 +0530
Change time	2026-09-08 11:23:31 +0530

1. What is metadata?

Metadata is the information about a file, like its size, owner, permissions, inode number and different timestamps.

2. Is metadata the same as file content?

No, metadata and file content are different. File content is the actual information inside the file, while metadata gives details about the file.

3. Which timestamp changes when file contents are modified?

The **Modification time (mtime)** changes when the contents of the file are changed.

```
anonymous@anonymous-supreme in ~/cyber_lab/reports
# ) cd ~/cyber_lab/reports

anonymous@anonymous-supreme in ~/cyber_lab/reports
# ) echo "This is a cybersecurity investigation report." > investigation.tx

anonymous@anonymous-supreme in ~/cyber_lab/reports
# ) ls -l investigation.txt
-rw-r--r-- 46 anonymous  8 Sep 11:23 investigation.txt

anonymous@anonymous-supreme in ~/cyber_lab/reports
# ) stat investigation.txt
File: investigation.txt
  Size: 46          Blocks: 8          IO Block: 4096   regular file
Device: 0,38      Inode: 5500882      Links: 1
Access: (0644/-rw-r--r--)  Uid: ( 1000/anonymous)   Gid: ( 1000/anonymous)
Access: 2026-09-07 01:18:44.628621728 +0530
Modify: 2026-09-08 11:23:31.434896487 +0530
Change: 2026-09-08 11:23:31.434896487 +0530
```

Exercise 11 – Explore `/etc`

Investigate:

/etc

..

Locate:

passwd

group

hostname

hosts

Note: Without modifying any system files, examine their contents where permitted.

Answer:

1. What kind of information is primarily stored under `/etc`?
2. What information does `/etc/passwd` contain?
3. What information does `/etc/hostname` contain?
4. Why is `/etc` security-sensitive?

What kind of information is primarily stored under /etc?

/etc mainly contains system configuration files and settings used by Linux and different services.

What information does /etc/passwd contain?

/etc/passwd contains information about user accounts, such as username, user ID, group ID, home directory and login shell.

What information does /etc/hostname contain?

/etc/hostname contains the hostname or name of the computer.

Why is /etc security-sensitive?

/etc is security-sensitive because it contains important system configuration and user-related files. Changing some of these files incorrectly or without permission can affect the system or create security problems.

```
anonymous@anonymous-supreme in ~/cyber_lab/reports
# ) cd /etc

anonymous@anonymous-supreme in /etc
# ) ls -l passwd group hostname hosts
-rw-r--r-- 1.4k root 4 Sep 16:43 group
-rw-r--r-- 18 root 27 Mar 2025 hostname
-rw-r--r-- 194 root 27 Mar 2025 hosts
-rw-r--r-- 2.9k root 4 Sep 16:41 passwd

anonymous@anonymous-supreme in /etc
# ) cat passwd cat group cat hostname cat hosts
root:x:0:0::/root:/usr/bin/bash
bin:x:1:1::/usr/bin/nologin
```

Exercise 12 – Explore Linux Log Storage

Log files found in /var/log

I found these log files:

- pacman.log
- lastlog
- snapper.log
- haskell-register.log
- Xorg.0.log
- Xorg.0.log.old
- firewalld
- wtmp
- wtmp.1
- btmp
- btmp.1

Log directories found

- audit/
- cups/
- garuda/
- gssproxy/
- i2pd/
- ipp-usb/
- journal/

- old/
- private/
- radius/
- samba/

Recently modified entries

Some recently modified entries were:

Entry	Modified
snapper.log	8 Sep 11:19
wtmp	8 Sep 09:54
firewalld	8 Sep 09:08
pacman.log	7 Sep 01:21
samba/	7 Sep 00:57

Logs I could read

As a normal user, I could read:

- pacman.log
- lastlog
- haskell-register.log
- Xorg.0.log
- Xorg.0.log.old
- wtmp
- wtmp.1

Some files like btmp, btmp.1, firewalld, and snapper.log were not readable as a normal user.

Answers

1. Why does Linux maintain logs?

Linux keeps logs to record different activities, errors, and events happening on the system.

2. Why are logs useful during incident investigation?

Logs help us find out what happened on the system and when it happened. They can also help us find suspicious activities.

3. Why might access to some logs be restricted?

Some logs contain important or sensitive information, so only authorized users should be able to access them.

Exercise 13 – Investigate /tmp

1. Files and directories currently present

There are both files and directories present in /tmp.

Some directories are:

- .font-unix
- .ICE-unix
- .X11-unix
- plasma-csd-generator.mTavRs
- plasmashell-dqWkAC
- Several systemd-private-* directories

Some files are:

- .X0-lock
- libinput-gestures-anonymous.lock
- libinput-gestures-anonymous.pid

There are also socket files such as plasmalogin--iwwFFO.

2. Hidden files

Yes, hidden files and directories are present. Examples include:

- .X0-lock
- .font-unix
- .ICE-unix
- .X11-unix
- .XIM-unix

They are hidden because their names start with ..

3. Ownership

I found entries owned by different users.

- /tmp is owned by root.
- plasma-csd-generator.mTavRs and plasmashell-dqWkAC are owned by anonymous.

- Some systemd-private-* directories are owned by root.
- plasmalogin--iwwFFO is owned by plasmalogin.

4. Recently modified entries

According to the timestamps, some recently modified entries were:

Entry	Modified
/tmp	8 Sep 11:29
plasma-csd-generator.mTavRs	8 Sep 09:09
plasmashell-dqWkAC	8 Sep 09:09
libinput-gestures-anonymous.lock	8 Sep 09:09
libinput-gestures-anonymous.pid	8 Sep 09:09

Answer

Why can temporary directories be relevant during a cybersecurity investigation?

Temporary directories can be useful during an investigation because programs and users may store temporary files there. An attacker may also use /tmp to store scripts, downloaded files or other temporary data. Therefore, checking /tmp can sometimes help find evidence of suspicious activity.

Exercise 14 – Explore the `/proc` Filesystem

Investigate:

```
/proc
```

```
...
```

Examine:

```
cat /proc/cpuinfo
```

```
cat /proc/meminfo
```

```
...
```

Then identify your current shell PID:

```
echo $$
```

```
...
```

Check whether a corresponding directory exists under `/proc`.

Answer:

1. What information does `/proc/cpuinfo` provide?
2. What information does `/proc/meminfo` provide?
3. Why do directories with numbers appear inside `/proc`?
4. Is `/proc` an ordinary disk directory?
5. Why could `/proc` be useful during security investigation?

1. What information does `/proc/cpuinfo` provide?

`/proc/cpuinfo` provides information about the CPU, such as the processor model, number of cores, CPU speed, cache size and supported features.

2. What information does `/proc/meminfo` provide?

`/proc/meminfo` provides information about the system's memory, such as total RAM, free memory, available memory and swap memory.

3. Why do directories with numbers appear inside `/proc`?

The numbered directories represent running processes. Each number is the PID of a process. For example, my shell had PID **33032**, so `/proc/33032` existed.

4. Is `/proc` an ordinary disk directory?

No. `/proc` is a virtual filesystem created by the Linux kernel. The information in it represents the current state of the system and processes.

5. Why could `/proc` be useful during security investigation?

`/proc` can be useful because it provides information about running processes, CPU, memory and other system details. It can help investigators identify unusual or suspicious processes running on the system

.

Exercise 15 – Explore `/dev`

/dev

...

Examine:

ls -l /dev/null

ls -l /dev/zero``

Answer:

1. What does `/dev` represent?
2. Are the entries ordinary user files?
3. Determine the file type of `/dev/null`.
4. Why does Linux expose devices through the filesystem?

What does /dev represent?

/dev contains files that represent devices connected to or used by the Linux system.

Are the entries ordinary user files?

No. Many entries in /dev are special device files used by the operating system to communicate with hardware or virtual devices.

Determine the file type of /dev/null.

/dev/null is a **character device file**. It is used to discard data written to it.

Why does Linux expose devices through the filesystem?

Linux uses the filesystem to provide a simple way for programs to communicate with devices. Programs can interact with devices using normal file operations like reading and writing.

Exercise 16 – Filesystem Space Investigation

Determine:

1. Total filesystem capacity.
2. Used space.
3. Available space.
4. Percentage of filesystem utilization.
5. Space occupied by your home directory.
6. Space occupied by the `cyber\_lab` directory.

Use appropriate combinations of:

`df`

`df -h`

`du`

`du -h`

`du -sh`

Answer:

> Explain the difference between `df` and `du`.

`df` is used to check the total, used, and available space of a filesystem.

`du` is used to check how much space a particular file or directory is using.

For example, `df -h /` shows the space of the filesystem, while `du -sh ~/cyber_lab` shows the space used by the `cyber_lab` folder.

```
anonymous@anonymous-supreme in /tmp
# ) df -h /
Filesystem      Size  Used Avail Use% Mounted on
/dev/nvme0n1p5 312G  135G  173G   44% /

anonymous@anonymous-supreme in /tmp
# ) du -sh ~
du: cannot read directory '/home/anonymous/.cache/yay/compiler-rt19/pkg': Permission denied
du: cannot read directory '/home/anonymous/.cache/yay/gcc14/pkg': Permission denied
du: cannot access '/home/anonymous/sample_dir/inside.txt': Permission denied
84K    /home/anonymous

anonymous@anonymous-supreme in /tmp took 7s176ms
[*] # du -sh ~/cyber_lab
4.0K   /home/anonymous/cyber_lab

anonymous@anonymous-supreme in /tmp
# ) du -h --max-depth=1 ~/cyber_lab
0      /home/anonymous/cyber_lab/evidence
0      /home/anonymous/cyber_lab/logs
0      /home/anonymous/cyber_lab/scripts
4.0K   /home/anonymous/cyber_lab/reports
4.0K   /home/anonymous/cyber_lab

anonymous@anonymous-supreme in /tmp
# ) _
mission denied
```

Exercise 17 – Cybersecurity Investigation Scenario

If suspicious activity is reported on a Linux system, I would investigate the following locations because each one can provide different types of security-related information.

Location	Why investigate?	Security-relevant information
/home	Contains files and folders of normal users.	User files, hidden files, scripts, downloads and suspicious programs.
/etc	Contains important system configuration files.	User information, hostname, services, network and authentication settings.
/var/log	Contains system and application logs.	Login activity, errors, firewall events and other system activities.
/tmp	Used for temporary files by applications and users.	Temporary scripts, downloaded files and possible suspicious activity.
/proc	Provides information about the running system.	Running processes, process IDs, memory and CPU information.
/dev	Contains special files representing devices.	Device information and access to hardware or virtual devices.
/root	Contains files belonging to the root user.	Administrator files, configuration files, scripts and command history.

Commands used

I would use commands that only examine the filesystem and do not modify or delete files.

```
ls -lah /home
```

```
ls -lah /etc
```

```
ls -lah /var/log
```

```
ls -lah /tmp
```

```
ls -lah /proc
```

```
ls -lah /dev
```

```
ls -lah /root
```

Other useful commands are:

```
find /home -maxdepth 2 -type f
```

```
find /tmp -maxdepth 2 -type f
```

```
cat /etc/hostname
```

```
cat /etc/passwd
```

```
ps aux
```

Observations

- /home can provide information about normal user activity and files.
- /etc contains system configuration and user-related information.
- /var/log can provide a timeline of system and security events.
- /tmp may contain temporary files or scripts created by applications or users.
- /proc shows information about running processes and system resources.
- /dev contains special device files used by Linux.
- /root may contain information related to administrator activity.

Conclusion

I would investigate all these locations because they can provide different types of evidence. /var/log, /home, /tmp, /etc, /proc, and /root would be especially useful for understanding user activity, system changes, logs, temporary files and running processes.

During the investigation, I would only examine the files and would not modify or delete any system files.